Medical Physics Department

Physiology Lab.

2nd Class

2026-2027

Lab.6

Clotting Time

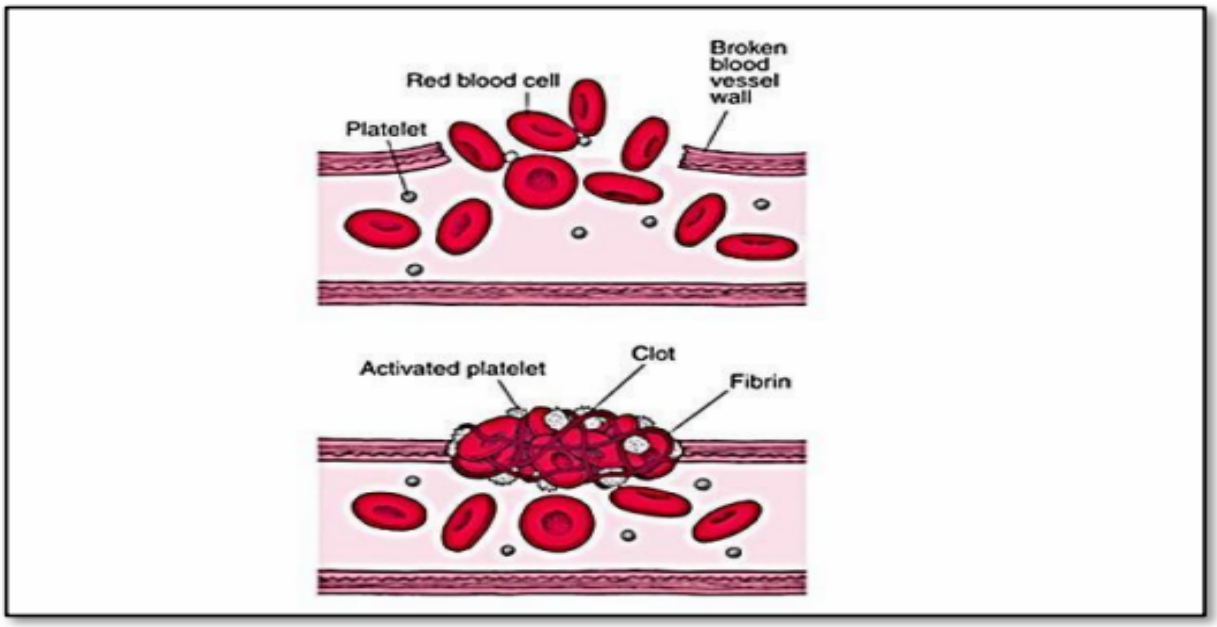


By:Dr.Hiba Khaleel Saeed

Clotting time: The time required for blood to form a clot, it test clot formation which depend on coagulation factors.

It is important for the blood to clot quickly when a vessel has been broken, but equally important for it not to clot in the absence of vessel damage.

The clotting time will be prolonged if the volume of blood per tube or the diameter of the tube is increased, or if the temperature decreased.



Clotting time is used for :

1. To diagnose and monitor the treatment of patients with a specific hereditary bleeding disorder (hemophilia).

2. Before surgical procedures.

3. For antitoxin therapy.

Methods

1. Capillary tube

2. Clotting Time using slide method
3. Clotting Time using Test Tube Method(.Lee and White Method).

Aim: To determine the clotting time of a subject.

Principle: A measure of the time required for blood to solidify (coagulate) after it has been removed from the body.

Material and instrument for clotting time test:

1.Fine capillary glass tubes of about 10 mm length

2.Lancet.

3.Stop watch.

4.Cotton and Alcohol 70%

Procedure:

1. Clean the finger with alcohol 70% and allow to dry.

2. Prick the finger by lancet.

3. Draw blood up in the capillary glass tube.

4. Start the stop watch.

5. After one minute start breaking small pieces of the

capillary tube every 30 second until a fibrin thread is

seen between the two broken ends.

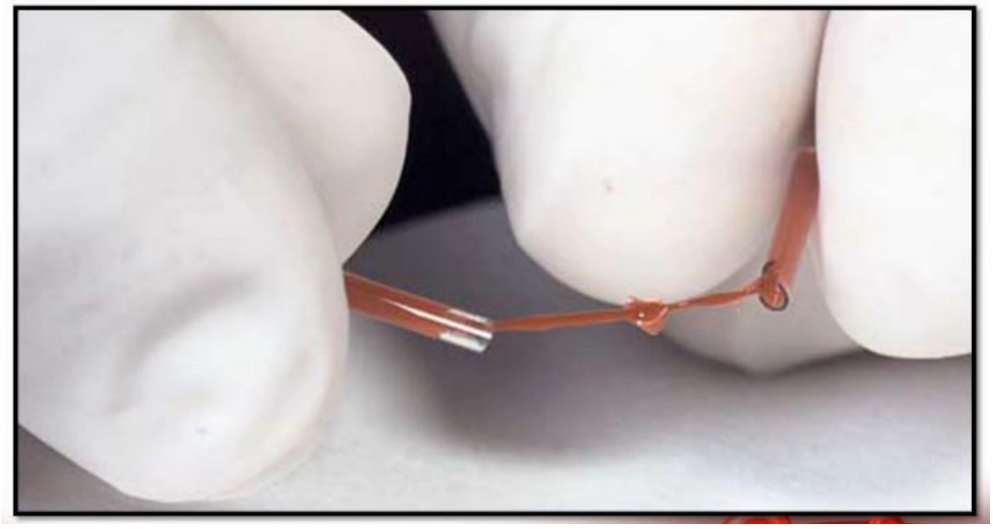


Calculating the clotting time by:

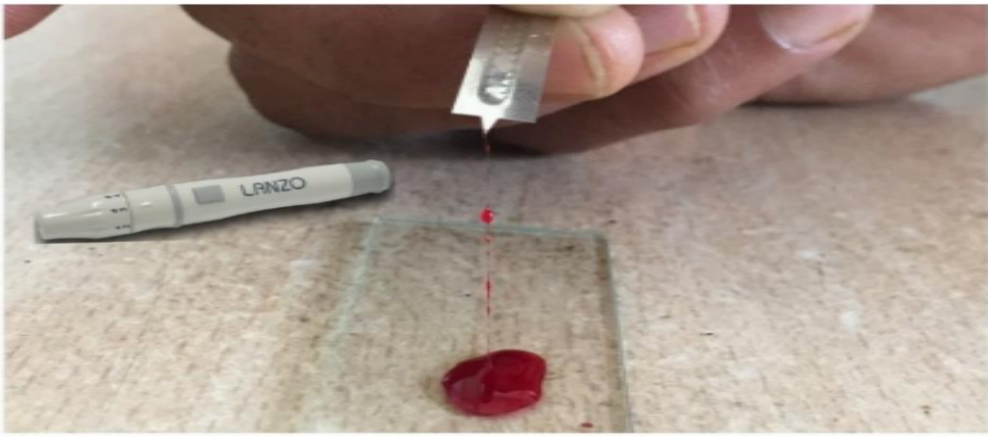
(The waiting time after the glass tube is filled+ no. of

capillary tubes breaks × 30 second)

Normal duration : 3 -8 minutes

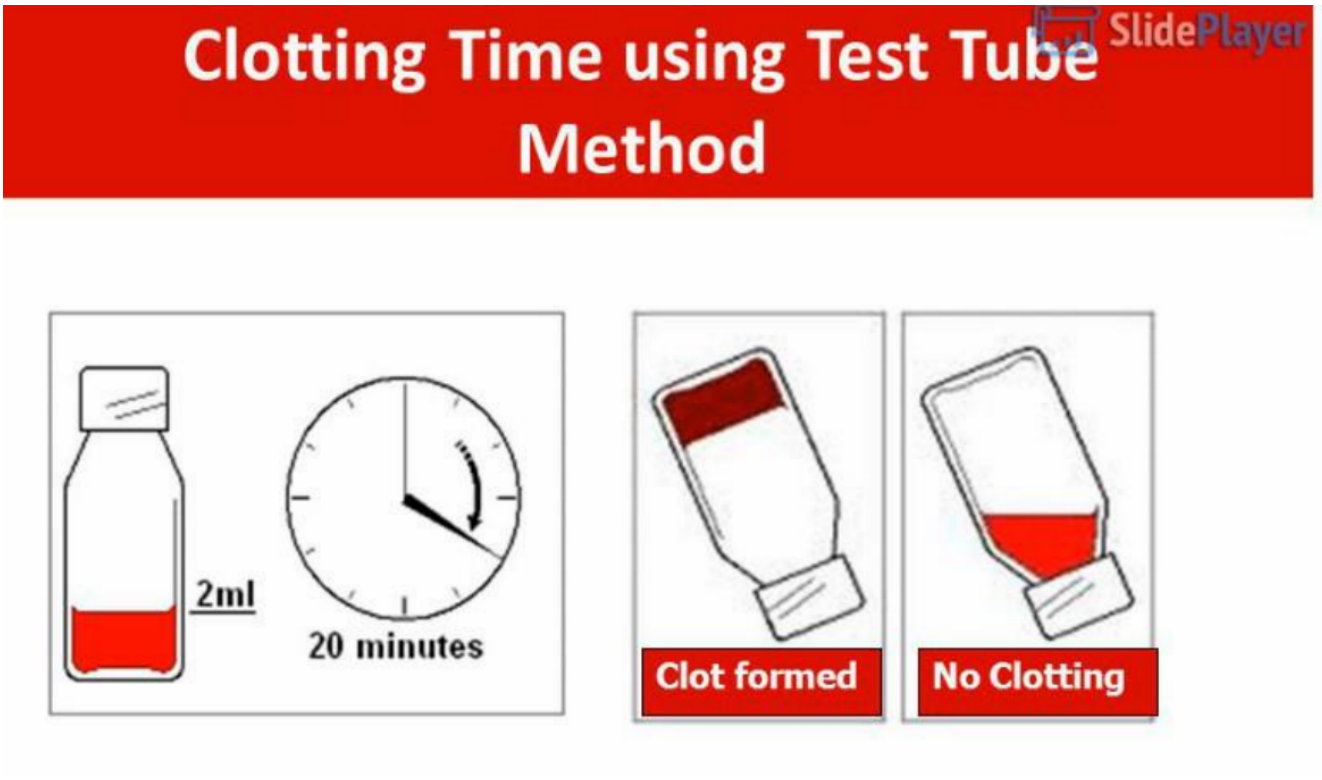


1. Clotting Time using slide method



3.Clotting Time using Test Tube Method(.Lee and White Method).

The test tube method for clotting time involves placing venous blood in a 37°C water bath and checking for clot formation at 30-second intervals by gently tilting the tubes. The time it takes for a clot to form in each tube is recorded, and the normal range is typically 6-10 minutes for this specific method. Another related test, using capillary tubes, involves breaking off small segments at 30-second intervals until a fibrin thread is visible between the broken ends.



Conditions where Clotting Time is Prolonged

1. Haemophilia (bleeding time is normal)
2. Christmas disease
3. Afibrinogenemia
4. Vitamin K deficiency
5. Liver disease
6. Anticoagulant therapy
7. Newborn baby.